

Ordinary Differential Equations

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September 7, 2026

Contents

1	First Order Differential Equations	2
1	First order linear differential equations	2
1.1	Homogeneous first order linear differential equation	4
1.2	Non-homogeneous first order linear differential equation	5
2	Separable equations	7
3	Population models	8
4	Exact equations and the range we can solve	9
4.1	$\mu = \mu(t)$	11
4.2	$\mu = \mu(y)$	11
4.3	Proof of theorem 1.4.2	12
5	The existence-uniqueness theorem: Picard iteration	13
5.1	Construction of Picard iterates	13
5.2	Convergence of Picard iterates	14
5.3	Is Picard iterates right	16
5.4	Uniqueness	18
5.5	Proof of (39)	20
	References	21

Chapter 1

First Order Differential Equations

1 First order linear differential equations

Main result of the section

In this section, we will introduce first order linear equation, including first order homogeneous linear equation and first order non-homogeneous linear equation. Also, we solve them as:

Theorem 1.1.1. *For equation in the form:*

$$\frac{dy}{dt} + a(t)y = 0,$$

the solution is

$$y(t) = c e^{-\int a(t) dt}$$

Moreover, if we have $y(t_0) = y_0$, the solution is

$$y(t) = y(t_0) \exp\left(\int_{t_0}^t a(s) ds\right)$$

and

Theorem 1.1.2. *For differential equations with the form:*

$$\frac{dy}{dt} + a(t)y = b(t) \quad (1)$$

the solution is

$$y = \frac{1}{\mu(t)} \left(\int \mu(t)b(t)dt + c \right), \quad \text{with } \mu(t) = \exp\left(\int a(t)dt\right).$$

Moreover, if we have $y_0 = y(t_0)$, then we have:

$$y = \frac{1}{\mu(t)} \left(\mu(t_0)y_0 + \int_{t_0}^t \mu(s)b(s)dt \right)$$

For the first order differential equations, we would consider to solve

$$\frac{dy}{dt} = f(t, y)$$

However, at least now, we are not able to solve this general form. We may consider some *linear* cases:

Definition 1.1.1 (First-order linear differential equation).

The general first-order linear differential equation is

$$\frac{dy}{dt} + a(t)y = b(t) \quad (2)$$

Needless to say, $a(t)$, $b(t)$ are assumed to be continuous functions of time. First, consider something simple, let's say: $b(t) = 0$ for all t .

Definition 1.1.2 (Homogeneous first-order linear differential equation).

The equation

$$\frac{dy}{dt} + a(t)y = 0 \quad (3)$$

is called homogeneous first-order linear differential equation

See section 1.1 for the solution.

Then, more difficult, we have:

Definition 1.1.3 (General first-order linear differential equation).

The equation

$$\frac{dy}{dt} + a(t)y = b(t) \quad (4)$$

is called non-homogeneous first-order linear differential equation

See section 1.2 for the solution.

1.1 Homogeneous first order linear differential equation

Let's consider:

$$\frac{dy}{dt} + a(t)y = 0.$$

Do some basic operation:

$$\frac{dy}{dt} = -a(t)y.$$

$$\frac{1}{y} \frac{dy}{dt} = -a(t).$$

(Yes, yes, there is a small problem: we've not yet proven that $y(t) \neq 0$ for all time. However, we will fix this gap later.)

Integrating both sides with respect to t gives

$$\int \frac{1}{y} \frac{dy}{dt} dt = - \int a(t) dt.$$

By the chain rule, $\frac{1}{y} \frac{dy}{dt} = \frac{d}{dt} \ln |y|$, hence

$$\ln |y(t)| = - \int a(t) dt + C,$$

where C is an arbitrary constant of integration. Exponentiating both sides yields

$$|y(t)| = e^C e^{-\int a(t) dt}.$$

Since $e^C > 0$, we may absorb the sign into a new constant $c = \pm e^C$, obtaining the general solution

$$\boxed{y(t) = c e^{-\int a(t) dt}} \quad (5)$$

where $c \in \mathbb{R}$ is an arbitrary constant. Note that the zero solution corresponds to $c = 0$.

Now we add the condition:

$$y(t_0) = y_0 \quad (6)$$

Then we integral (5) between t_0 and t .

Recalling $\frac{1}{y} \frac{dy}{dt} = -a(t)$, we could write:

$$\int_{t_0}^t \frac{d}{ds} \ln |y(s)| ds = \int_{t_0}^t -a(s) ds$$

So

$$\ln |y(t)| - \ln |y(t_0)| = \ln \left| \frac{y(t)}{y(t_0)} \right| = - \int_{t_0}^t a(s) ds$$

Thus,

$$\left| \frac{y(t)}{y(t_0)} \right| = \exp\left(- \int_{t_0}^t a(s) ds\right)$$

The problem now becomes +1 or -1. By taking $t = t_0$, we can find it's +1.

So

$$y(t) = y(t_0) \exp\left(\int_{t_0}^t a(s) ds\right) \quad (7)$$

1.2 Non-homogeneous first order linear differential equation

Now we consider

$$\frac{dy}{dt} + a(t)y = b(t) \quad (8)$$

There is a method called *variance of constant*. Roughly speaking, we may use this method to all non-homogeneous linear differential equations.

Let's introduce it:

For

$$\frac{dy}{dt} + a(t)y = 0 \quad (9)$$

our solution is

$$y(t) = c \exp\left[-\int a(t) dt\right] \quad (10)$$

So for (8), we may consider the solution as:

$$y(t) = c(t) \exp\left(-\int a(t) dt\right) \quad (11)$$

And now our goal is to find $c(t)$.

$$y(t) = c(t) \exp\left(-\int a(t) dt\right) \quad (12)$$

$\Downarrow$

$$\frac{dy}{dt} = c'(t) \exp\left(-\int a(t) dt\right) - \underbrace{c(t)a(t) \exp\left(-\int a(t) dt\right)}_{=y(t)} \quad (13)$$

So

$$\begin{aligned} \frac{dy}{dt} + a(t)y &= c'(t) \exp\left(-\int a(t) dt\right) \\ &= b(t) \end{aligned}$$

So

$$c'(t) = b(t) \exp\left(\int a(t) dt\right) \quad (14)$$

$$c(t) = \int \left(b(t) \exp\left(\int a(t) dt\right) \right) dt \quad (15)$$

And

$$y(t) = \exp\left(-\int a(t) dt\right) \left(\int b(t) \exp\left(\int a(t) dt\right) dt + c \right) \quad (16)$$

2 Separable equations

Recall how we deal with linear differential equation:

$$\frac{1}{y(t)} \frac{dy}{dt} = -a(t)$$

$\Downarrow$

$$\frac{d}{dt} \ln |g(t)| = -a(t)$$

So for a more general ODE with the form:

$$\frac{dy}{dt} = \frac{g(t)}{f(y)} \tag{17}$$

We could write it as

$$\frac{d}{dt} F(y(t)) = g(t), \quad \text{with } F(y) = \int f(y) dy$$

So the answer is:

$$F(y(t)) = \int g(t) dt + c \tag{18}$$

We could also add an assumption $y(t_0) = y_0$

So

$$F(y(t)) - F(y(t_0)) = \int_{t_0}^t g(s) ds$$

$$\int_{y_0}^y f(r) dr = \int_{t_0}^t g(s) ds$$

3 Population models

Let $p(t)$ denotes the population of a given species at time t , $r(t, p)$ denotes the difference between its birth rate and death rate.

For the simplest, we think that r will not change with t and p , which means it's a constant. The simplest model:

$$\frac{dp(t)}{dt} = ap(t), \quad a \text{ is a constant}$$

Also, assume $p(t_0) = p_0$

We can easily find the solution

$$p(t) = p_0 \exp(a(t - t_0)) \quad (19)$$

This result works very well if the population is small. However, if the population grows larger and larger, it falls. So we can let

$$\frac{dr}{dt} = ar - br^2 \quad (20)$$

It's also a sparable problem, though the integer part is a bit complex (but absolutly solvable!)

The answer is:

$$p(t) = \frac{ap_0 e^{a(t-t_0)}}{a - bp_0 + bp_0 e^{a(t-t_0)}} = \frac{ap_0}{bp_0 + (a - bp_0)e^{-a(t-t_0)}}.$$

4 Exact equations and the range we can solve

Consider all differential equation we've solved, we can put them in this form:

$$\frac{d}{dt}\phi(t, y) = 0 \quad (21)$$

Example 1.4.1 (ϕ for first order linear differential equations).

Thinking about all separable differential equation (it would also contains all first order linear differential equations) we've solved before:

$$\frac{dy}{dt} = g(t)h(y(t))$$

We can all write this equation in this form:

$$\phi(t, y) = \int^y \frac{1}{h(s)} ds - \int^t g(r) dr$$

More generally, we can solve all differential equations which can be written in this form:

$$\frac{d}{dt}\phi(t, y(t)) = 0 \quad (22)$$

Remark 4.1. Actually, 'we can solve' doesn't mean that we can get explicit solution. It just become a equation without differential term. But with the aid of computer, we can solve its numer with any desired accuracy.

Consider:

$$\frac{d}{dt}\phi(t, y) = \frac{\partial \phi}{\partial t} + \frac{\partial \phi}{\partial y} \frac{dy}{dt} \quad (23)$$

One natural question is that for all $M(t, y), N(t, y)$, does there exist ϕ s.t.

$$M = \frac{\partial \phi}{\partial t} \quad N = \frac{\partial \phi}{\partial y} \quad (24)$$

The answer is **NO**.

Theorem 1.4.2. *Let $M(t, y)$ and $N(t, y)$ be continuous and have continuous partial derivatives with respect to t and y in the rectangle R consisting of those points (t, y) with $a < t < b$ and $c < y < d$. There exists a function $\phi(t, y)$ such that*

$$M(t, y) = \frac{\partial \phi}{\partial t} \quad \text{and} \quad N(t, y) = \frac{\partial \phi}{\partial y}$$

if, and only if,

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}$$

in R .

The entire proof will be in 4.3.

Definition 1.4.1 (Exact equations).

The differential equation

$$M(t, y) + N(t, y) \frac{dy}{dt} = 0$$

is said to be exact if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}.$$

For exact equations, we can write:

$$\phi(t, y) = \int M(t, y) dy + \int \left(N(t, y) - \int \frac{\partial M(t, y)}{\partial y} dt \right) dy \quad (25)$$

Remark 4.3. It's not necessary that $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}$ in a rectangle R as theorem 1.4.2 states. It can be any region without 'hole' inside.

Remark 4.4. For differential equation $dy/dt = f(t, y)$, it can be written in the form $M(t, y) + N(t, y)(dy/dt) = 0$ by setting $M(t, y) = -f(t, y)$ and $N(t, y) = 1$.

Now we've solved Exact equations. Suppose

$$M(t, y) + N(t, y) \frac{dy}{dt} = 0 \quad (26)$$

is not exact. Could we solve it?

A natural idea is to multiply both terms with a function $\mu(t, y)$:

$$M(t, y)\mu(t, y) + \mu(t, y)N(t, y) \frac{dy}{dt} = 0 \quad (27)$$

And if μ is well-chosen, it could become an exact equation, and we can have

$$\frac{\partial}{\partial y} \left(M(t, y)\mu(t, y) \right) = \frac{\partial}{\partial t} \left(N(t, y)\mu(t, y) \right) \quad (28)$$

Definition 1.4.2 (Integrating Factor).

Any function $\mu(t, y)$ satisfying (28) is called integrating factor for the equation (26)

Then we try to find μ .

$$\frac{\partial}{\partial y} \left(M(t, y)\mu(t, y) \right) = \frac{\partial}{\partial t} \left(N(t, y)\mu(t, y) \right)$$

$\Downarrow$

$$M \frac{d}{dy} \mu + \mu \frac{d}{dy} M = N \frac{d}{dt} \mu + \mu \frac{d}{dt} N$$

$\Downarrow$

$$-N \frac{d\mu}{dt} + M \frac{d\mu}{dy} = \left(\frac{dN}{dt} - \frac{dM}{dy} \right) \mu$$

Solving a general $\mu(t, y)$ is almost as difficult as solving the ODE equation.

So if μ could only depend on t or y , we could be able to solve it. **It's just an assumption.**

We cannot say that there must exist $\mu = \mu(t)$ or $\mu = \mu(y)$. Just for some special cases, we calculate some μ below and then we find if we are lucky enough to obtain $\mu = \mu(t)$ or $\mu = \mu(y)$.

4.1 $\mu = \mu(t)$

If μ depends only on t , then $\partial\mu/\partial y = 0$. Therefore,

$$-N \frac{d\mu}{dt} = \left(\frac{\partial N}{\partial t} - \frac{\partial M}{\partial y} \right) \mu,$$

and hence

$$\frac{1}{\mu} \frac{d\mu}{dt} = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial t}}{N}. \quad (29)$$

Since the left-hand side depends only on t , the expression on the right-hand side must also be a function of t alone. If

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial t}}{N} = f(t),$$

then integrating (29) gives

$$\boxed{\mu(t) = C \exp \left(\int f(t) dt \right)}, \quad C \neq 0. \quad (30)$$

4.2 $\mu = \mu(y)$

If μ depends only on y , then $\partial\mu/\partial t = 0$. Therefore,

$$M \frac{d\mu}{dy} = \left(\frac{\partial N}{\partial t} - \frac{\partial M}{\partial y} \right) \mu,$$

and hence

$$\frac{1}{\mu} \frac{d\mu}{dy} = \frac{\frac{\partial N}{\partial t} - \frac{\partial M}{\partial y}}{M}. \quad (31)$$

Since the left-hand side depends only on y , the expression on the right-hand side must also be a function of y alone. If

$$\frac{\frac{\partial N}{\partial t} - \frac{\partial M}{\partial y}}{M} = g(y),$$

then integrating (31) gives

$$\mu(y) = C \exp \left(\int g(y) dy \right), \quad C \neq 0. \quad (32)$$

4.3 Proof of theorem 1.4.2

Proof. Fix $t_0 \in (a, b)$. First, suppose that ϕ satisfies

$$\frac{\partial \phi}{\partial t} = M(t, y).$$

Integrating with respect to t , while regarding y as a constant, gives

$$\phi(t, y) = \int_{t_0}^t M(s, y) ds + h(y), \text{ where } h(y) \text{ is an arbitrary function depending only on } y. \quad (33)$$

Differentiating (33) with respect to y , we obtain

$$\frac{\partial \phi}{\partial y} = \int_{t_0}^t \frac{\partial M}{\partial y}(s, y) ds + \frac{dh}{dy}(y).$$

We also require $\partial \phi / \partial y = N(t, y)$. Therefore,

$$\frac{dh}{dy}(y) = N(t, y) - \int_{t_0}^t \frac{\partial M}{\partial y}(s, y) ds. \quad (34)$$

The left-hand side of (34) depends only on y , so the right-hand side must also be independent of t . Differentiating the right-hand side with respect to t , we find

$$\frac{\partial}{\partial t} \left(N(t, y) - \int_{t_0}^t \frac{\partial M}{\partial y}(s, y) ds \right) = \frac{\partial N}{\partial t}(t, y) - \frac{\partial M}{\partial y}(t, y) = 0$$

Thus, if such a function ϕ exists, then necessarily

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}.$$

Conversely, if this equality holds throughout R , then the right-hand side of (34) has zero derivative with respect to t , and hence depends only on y . We can therefore integrate it with respect to y to obtain a function $h(y)$. Substituting this h into (33) gives a function satisfying

$$\frac{\partial \phi}{\partial t} = M(t, y) \quad \text{and} \quad \frac{\partial \phi}{\partial y} = N(t, y).$$

Consequently, such a function ϕ exists if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}.$$

□

5 The existence-uniqueness theorem: Picard iteration

Many first order differential equations (actually, most) we cannot get the explicit solution. However, can we prove that there exists a solution? If so, could we prove that it's the only solution, or it's the only solution under some conditions?

Consider

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0 \quad (35)$$

So

$$y(t) = y_0 + \int_{t_0}^t f(s, y(s)) ds \quad \text{as } y(t) \text{ is continuous and } y(t_0) = y_0 \quad (36)$$

Define an operator¹

$$\begin{aligned} \mathcal{L}[y](t) &= y_0 + \int_{t_0}^t f(s, y(s)) ds \\ \mathcal{L} : C([0, T]) &\rightarrow C([0, T]) \end{aligned}$$

For equation

$$y(t) = \mathcal{L}[y](t)$$

we call it *fixed point* of $\mathcal{L}$.

So the problem: Finding the solution of ODE $\rightarrow$ Finding the fixed point of $\mathcal{L}$. We'll use a sequence of $\{y_n\}$ to approximate.

In the following subsections, we will learn how to construct a sequence of $\{y_n\}$ in 5.1. We'll talk about whether the sequence will converge in 5.2 (But we don't care where it converges). We'll prove that the convergence result is the solution of (36) in 5.3. Last, we'll prove the uniqueness of the solution in 5.4

5.1 Construction of Picard iterates

To solve

$$y(t) = y_0 + \int_{t_0}^t f(s, y(s)) ds \quad \text{as } y(t) \text{ is continuous and } y(t_0) = y_0 \quad (37)$$

we create a sequence of $\{y(t)\}$ to approximate the answer. We start with the constant function

$$y_0(t) = y_0.$$

Substituting this initial guess into the right-hand side of the integral equation gives the first Picard iterate:

$$y_1(t) = \mathcal{L}[y_0](t) = y_0 + \int_{t_0}^t f(s, y_0(s)) ds = y_0 + \int_{t_0}^t f(s, y_0) ds.$$

¹Actually I still don't get fully understand why we do so complex here. It may have some connection with the space? Anyway, see it, accept it, and see what's going on later.

If $y_1(t) = y_0(t)$, then y_0 is already a fixed point of $\mathcal{L}$, so it is a solution of the initial value problem. If not, we substitute y_1 into $\mathcal{L}$ again and obtain

$$y_2(t) = \mathcal{L}[y_1](t) = y_0 + \int_{t_0}^t f(s, y_1(s)) ds.$$

We continue this process. At each step, if

$$y_{n+1}(t) = y_n(t),$$

then y_n is a fixed point of $\mathcal{L}$ and hence a solution. Otherwise, we use y_{n+1} to construct the next iterate. In general, the Picard iterates are defined by

$$y_0(t) = y_0, \quad y_{n+1}(t) = y_0 + \int_{t_0}^t f(s, y_n(s)) ds, \quad n \geq 0. \quad (38)$$

Notice that every iterate satisfies $y_n(t_0) = y_0$.

Then we'll show the convergence of Picard iterates.

5.2 Convergence of Picard iterates

Lemma 5.1. Choose any two positive numbers a and b , and let R be the rectangle:

$$t_0 \leq t \leq t_0 + a, \quad |y - y_0| \leq b.$$

Compute

$$M = \max_{(t,y) \in R} |f(t, y)|, \quad \text{and set } \alpha = \min\left(a, \frac{b}{M}\right).$$

Then,

$$|y_n(t) - y_0| \leq M(t - t_0)$$

for $t_0 \leq t \leq t_0 + \alpha$.

Proof. For $n = 0$,

$$|y_0(t) - y_0| = 0 \leq M(t - t_0).$$

Suppose the result is true for $n = j$. Since $t - t_0 \leq \alpha$,

$$|y_j(t) - y_0| \leq M(t - t_0) \leq M\alpha \leq b.$$

Thus, $(t, y_j(t)) \in R$, so $|f(t, y_j(t))| \leq M$. Therefore,

$$\begin{aligned} |y_{j+1}(t) - y_0| &= \left| \int_{t_0}^t f(s, y_j(s)) ds \right| \\ &\leq \int_{t_0}^t |f(s, y_j(s))| ds \\ &\leq M(t - t_0). \end{aligned}$$

Hence the result follows for every $n \geq 0$ by induction. $\square$

We now show the convergence of $\{y_n\}$. First, we write y_n in this form:

$$y_n(t) = y_0(t) + [y_1(t) - y_0(t)] + \cdots + [y_n(t) - y_{n-1}(t)]$$

So the sequence $y_n(t)$ converges iff the infinite series converges:

$$[y_1(t) - y_0(t)] + [y_2(t) - y_1(t)] + \cdots + [y_n(t) - y_{n-1}(t)] + \cdots$$

Then, we try to prove a stronger result:

$$\sum_{n=1}^{\infty} |y_n(t) - y_{n-1}(t)| < \infty \Rightarrow \sum_{n=1}^{\infty} [y_n(t) - y_{n-1}(t)] \text{ converges.} \quad (39)$$

This is not obvious (at least not obvious for me). The result used in (39) will be proved in 5.5.

$$L = \max_{(t,y) \in R} \left| \frac{\partial f}{\partial y}(t, y) \right|.$$

By the mean value theorem, for some $\xi_n(s)$ between $y_n(s)$ and $y_{n-1}(s)$,

$$\begin{aligned} |f(s, y_n(s)) - f(s, y_{n-1}(s))| &= \left| \frac{\partial f}{\partial y}(s, \xi_n(s)) \right| |y_n(s) - y_{n-1}(s)| \\ &\leq L |y_n(s) - y_{n-1}(s)|. \end{aligned}$$

where

$$L = \max_{(t,y) \in R} \left| \frac{\partial f}{\partial y}(t, y) \right|.$$

First,

$$\begin{aligned} |y_1(t) - y_0(t)| &= \left| \int_{t_0}^t f(s, y_0) ds \right| \\ &\leq M(t - t_0). \end{aligned}$$

Then

$$\begin{aligned} |y_2(t) - y_1(t)| &= \left| \int_{t_0}^t [f(s, y_1(s)) - f(s, y_0(s))] ds \right| \\ &\leq L \int_{t_0}^t |y_1(s) - y_0(s)| ds \\ &\leq ML \int_{t_0}^t (s - t_0) ds \\ &= \frac{ML(t - t_0)^2}{2!}, \end{aligned}$$

and

$$\begin{aligned} |y_3(t) - y_2(t)| &\leq L \int_{t_0}^t |y_2(s) - y_1(s)| ds \\ &\leq \frac{ML^2(t - t_0)^3}{3!}. \end{aligned}$$

Continuing in the same way,

$$\boxed{|y_n(t) - y_{n-1}(t)| \leq \frac{ML^{n-1}(t - t_0)^n}{n!}, \quad n \geq 1.} \quad (40)$$

If $L = 0$, then $y_2 = y_1$. For $L > 0$ and $t_0 \leq t \leq t_0 + \alpha$,

$$\begin{aligned} \sum_{n=1}^{\infty} |y_n(t) - y_{n-1}(t)| &\leq \sum_{n=1}^{\infty} \frac{ML^{n-1}(t - t_0)^n}{n!} \\ &= \frac{M}{L} \sum_{n=1}^{\infty} \frac{[L(t - t_0)]^n}{n!} \\ &= \frac{M}{L} (e^{L(t-t_0)} - 1) \\ &\leq \boxed{\frac{M}{L} (e^{\alpha L} - 1)} < \infty. \end{aligned}$$

5.3 Is Picard iterates right

Till now, we've proved that $\{y_n\}$ converges. However, we still don't know whether $\lim_{n \rightarrow \infty} y_n(t)$ is the answer of (36).

Consider $\lim_{n \rightarrow \infty} y_n(t) = y$, and we take limit of both sides of:

$$y_{n+1}(t) = y_0 + \int_{t_0}^t f(s, y_n(s)) ds$$

Then we get

$$y = y_0 + \lim_{n \rightarrow \infty} \int_{t_0}^t f(s, y_n(s)) ds \quad (41)$$

Then what we need to do is to justify the right hand side equals to $\int_{t_0}^t f(s, y(s)) ds$.

One could also understand that the limit could *pass* the integral, as:

$$\lim_{n \rightarrow \infty} \int_{t_0}^t f(s, y_n(s)) ds = \int_{t_0}^t \left(\lim_{n \rightarrow \infty} f(s, y_n(s)) \right) ds = \int_{t_0}^t f(s, y(s)) ds$$

We still need to justify this step. For $m > n$, by (40),

$$\begin{aligned} |y_m(t) - y_n(t)| &\leq \sum_{k=n+1}^m |y_k(t) - y_{k-1}(t)| \\ &\leq \sum_{k=n+1}^m \frac{ML^{k-1}(t-t_0)^k}{k!} \\ &\leq \sum_{k=n+1}^m \frac{ML^{k-1}\alpha^k}{k!}. \end{aligned}$$

Let $m \rightarrow \infty$. Then

$$\sup_{t_0 \leq t \leq t_0 + \alpha} |y(t) - y_n(t)| \leq \sum_{k=n+1}^{\infty} \frac{ML^{k-1}\alpha^k}{k!} \rightarrow 0. \quad (42)$$

Thus,

$$y_n \rightarrow y \quad \text{uniformly on } [t_0, t_0 + \alpha].$$

By the mean value theorem,

$$\begin{aligned} |f(s, y_n(s)) - f(s, y(s))| &\leq L|y_n(s) - y(s)| \\ &\leq L \sup_{t_0 \leq r \leq t_0 + \alpha} |y_n(r) - y(r)|. \end{aligned}$$

Therefore,

$$\begin{aligned} &\left| \int_{t_0}^t f(s, y_n(s)) ds - \int_{t_0}^t f(s, y(s)) ds \right| \\ &\leq \int_{t_0}^t |f(s, y_n(s)) - f(s, y(s))| ds \\ &\leq \alpha L \sup_{t_0 \leq r \leq t_0 + \alpha} |y_n(r) - y(r)| \\ &\rightarrow 0. \end{aligned}$$

So the limit can pass through the integral:

$$\lim_{n \rightarrow \infty} \int_{t_0}^t f(s, y_n(s)) ds = \int_{t_0}^t f(s, y(s)) ds.$$

Taking the limit in the Picard iteration gives

$$\boxed{y(t) = y_0 + \int_{t_0}^t f(s, y(s)) ds.} \quad (43)$$

We also need to show that $y(t)$ is continuous. First,

$$y_0(t) = y_0 \implies y_0 \text{ is continuous.}$$

If y_n is continuous, then

$$s \mapsto f(s, y_n(s)) \text{ is continuous}$$

and

$$y_{n+1}(t) = y_0 + \int_{t_0}^t f(s, y_n(s)) ds \implies y_{n+1} \text{ is continuous.}$$

Thus, every y_n is continuous. Fix $t^* \in [t_0, t_0 + \alpha]$. For every $\varepsilon > 0$, the uniform convergence in (42) gives an N such that

$$\sup_{t_0 \leq t \leq t_0 + \alpha} |y(t) - y_N(t)| < \frac{\varepsilon}{3}.$$

Since y_N is continuous, there exists $\delta > 0$ such that

$$|t - t^*| < \delta \implies |y_N(t) - y_N(t^*)| < \frac{\varepsilon}{3}.$$

Therefore,

$$\begin{aligned} |y(t) - y(t^*)| &\leq |y(t) - y_N(t)| + |y_N(t) - y_N(t^*)| + |y_N(t^*) - y(t^*)| \\ &< \varepsilon. \end{aligned}$$

Hence,

$$\boxed{y \in C([t_0, t_0 + \alpha])}. \quad (44)$$

Finally,

$$y(t_0) = y_0, \quad \frac{dy}{dt} = f(t, y(t)).$$

Thus, $y(t)$ is a solution of (36).

5.4 Uniqueness

Theorem 1.5.2. *Let f and $\partial f / \partial y$ be continuous in the rectangle*

$$R : \quad t_0 \leq t \leq t_0 + a, \quad |y - y_0| \leq b.$$

Compute

$$M = \max_{(t,y) \in R} |f(t, y)|, \quad \text{and set } \alpha = \min \left(a, \frac{b}{M} \right).$$

Then, the initial-value problem

$$y' = f(t, y), \quad y(t_0) = y_0 \quad (45)$$

has a unique solution $y(t)$ on the interval $t_0 \leq t \leq t_0 + \alpha$. In other words, if $y(t)$ and $z(t)$ are two solutions of (45), then $y(t)$ must equal $z(t)$ for $t_0 \leq t \leq t_0 + \alpha$.

Proof. Theorem 2 guarantees the existence of at least one solution $y(t)$ of (45). Sup-

pose that $z(t)$ is a second solution of (45). Then,

$$y(t) = y_0 + \int_{t_0}^t f(s, y(s)) ds \quad \text{and} \quad z(t) = y_0 + \int_{t_0}^t f(s, z(s)) ds.$$

Subtracting these two equations gives

$$\begin{aligned} |y(t) - z(t)| &= \left| \int_{t_0}^t [f(s, y(s)) - f(s, z(s))] ds \right| \\ &\leq \int_{t_0}^t |f(s, y(s)) - f(s, z(s))| ds \\ &\leq L \int_{t_0}^t |y(s) - z(s)| ds, \end{aligned}$$

where L is the maximum value of $|\partial f / \partial y|$ for (t, y) in R . As Lemma 5.3 below shows, this inequality implies that $y(t) = z(t)$. Hence, the initial-value problem (45) has a unique solution $y(t)$. $\square$

Lemma 5.3. *Let $w(t)$ be a nonnegative function, with*

$$w(t) \leq L \int_{t_0}^t w(s) ds.$$

Then, $w(t)$ is identically zero.

Proof. Set

$$U(t) = \int_{t_0}^t w(s) ds.$$

Then

$$U'(t) = w(t) \leq L \int_{t_0}^t w(s) ds = LU(t).$$

Therefore,

$$\begin{aligned} \frac{d}{dt} \left(e^{-L(t-t_0)} U(t) \right) &= e^{-L(t-t_0)} (U'(t) - LU(t)) \\ &\leq 0. \end{aligned}$$

Thus,

$$0 \leq e^{-L(t-t_0)} U(t) \leq U(t_0) = 0,$$

so $U(t) = 0$. Finally,

$$0 \leq w(t) \leq L \int_{t_0}^t w(s) ds = LU(t) = 0.$$

Hence, $w(t) = 0$ for $t_0 \leq t \leq t_0 + \alpha$. $\square$

5.5 Proof of (39)

First, consider a sequence $\{a_n\}_{n \geq 1} \subset \mathbb{R}$ such that

$$\sum_{n=1}^{\infty} |a_n| < \infty.$$

Define

$$S_n = \sum_{k=1}^n a_k, \quad A_n = \sum_{k=1}^n |a_k|.$$

Since $\{A_n\}$ converges, it is a Cauchy sequence. Thus,

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad m > n \geq N$$

$$\Downarrow$$

$$|A_m - A_n| = \sum_{k=n+1}^m |a_k| < \varepsilon.$$

Then

$$\begin{aligned} |S_m - S_n| &= \left| \sum_{k=n+1}^m a_k \right| \\ &\leq \sum_{k=n+1}^m |a_k| \\ &< \varepsilon. \end{aligned}$$

Therefore,

$$\{S_n\} \text{ is Cauchy} \xrightarrow{\mathbb{R} \text{ is complete}} S_n \longrightarrow S \in \mathbb{R},$$

and hence

$$\boxed{\sum_{n=1}^{\infty} |a_n| < \infty \implies \sum_{n=1}^{\infty} a_n \text{ converges.}} \quad (46)$$

References

- [1] Martin Braun, *Differential Equations and Their Applications: An Introduction to Applied Mathematics*, 4th ed., Texts in Applied Mathematics, vol. 11, Springer-Verlag, New York, 1993. doi:10.1007/978-1-4612-4860-1.